WASTEWISE-ML · PROJECT PLAN

WasteWise-ml project plan

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DOCUMENT CONTROL

Document ID	WML-PLN-001	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/11-plan.md @ 9031e25		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First project plan (kit 1.8 and 1.9.1 upgrade), stage P3 reached; P4 to P6 on hold

WasteWise-ml project plan

WasteWise-ml is at TRL 3, stage P3 (design outputs): the model is sized, the scanning rig is constructable and detailed to D3, and the pipeline documents exist. The next gate is P4 (verification), which needs a field partner, field photos and the evaluation set, and is on hold while the portfolio cap is TRL 3 and stage P3. **Decisions requested of Amish:** the concept selection D1 (option A recommended, WML-DDR-004) and the lens height D2 (420 mm recommended, WML-DEC-001). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Requirements and pipeline	Project lead	PRB, STM, OPT, URS, PRS, RTM, REG, FEA, RSK, MKT, PLN	Done at P3 (2026-10-25); STEM check open on eight rows (S2, M2, M4, M5, B2, C3, E4, OB5)
Scanning rig	Design lead	Model at D3, drawing WML-DWG-001 Rev P3, making sketches WML-DWG-101 to 107, build plan WML-BLD-001	Done on paper; arm and tray wait for the bought phone (confirm items 1 and 2)
Model and app	ML lead	Taxonomy, model card template, per-site grade file, app result screen and voice (F1)	Taxonomy done; app and training not started (TRL 4)
Data and consent	Data lead with the partner	Data protocol, consent form, labeling plan, 10,000 field photos, 2,000-image evaluation set	Not started (TRL 4, on hold)
Partner and co-design	Project lead	Agreement with the first candidate partner (SWaCH, Pune), co-design sessions, buyers' grade list	Not started (on hold)
Photoreal renders	Coordinator (Amish's Mac)	`media/options.png`, product renders from `cad/src/product_model.py`, cards	Scenes ready; renders pending

Resources needed

Table 2. Resources (planning figures from the BOM and WML-CAL-001; the labeling rate is set with the partner, D7).

Resource	Deliverable	Effort	Estimated cost (USD)
Partner members, labeling	10,000 labeled field photos	About 72 h	430 (BOM line 8)
Partner members and buyers, ground truth	2,000-image evaluation set with 400 hazards	About 43 h	260 (BOM line 9)
Cloud GPU	Fine-tuning and evaluation	About 40 GPU hours	40 (BOM line 10)
Two pilot stations	Hardware for two sites	About a day's workshop time each	626.78 (2 x 313.39)
App developer	Android app: capture, result screen, voice, shutter button, grade file, consent and upload queue	To estimate	To assess
Regulatory reviewer	DPDP protocol review, legal metrology, battery and bio-medical waste routes (REG G2 to G6)	To estimate	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Done (2026-10-25)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT, REG complete	Done (2026-10-25)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed, FEA closed	Done except the STEM check (eight rows open: field data, trials on bought parts, and D2) and the concept choice (D1, awaiting Amish)
P3 Design outputs	Model at D3, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability complete	Done (2026-10-25)
P4 Verification	Field evaluation report (PR-001 to PR-004, PR-011), timed and usability trials (PR-007, PR-008, PR-016, PR-022), rig first checks	On hold (TRL 4)
P5 Validation	Pilot with the partner at two sites; consent protocol in use	On hold
P6 Commercialization	Not planned: an open, non-commercial route (WML-MKT-001)	On hold

Next steps and decision requested

1. **Amish:** decide the concept selection D1 (option A, the current design, recommended; WML-DDR-004) and the lens height D2 (420 mm recommended; WML-DEC-001).

2. Coordinator: render `media/options.png` and the product renders named in `cad/src/product_model.py` on the Mac.
3. When TRL 4 is opened: approach SWaCH, write the data protocol with them, buy and measure the reference phone and shutter button (confirm items 1, 2 and 5), then build one rig to WML-BLD-001.
4. Measure the learning curve on the first 2,000 field photos before collecting the rest (closes STEM M4 and M5 or changes the plan).

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